

CSCE 676 DATA MINING · MODULE 01 · LECTURE NOTES

Two Problems, Not One

Finding structure, and deciding whether it is there.

MODULE OUTCOMES

- State the two problems and why both are necessary.
- Explain what scale does to algorithm choice.
- Explain why search guarantees a finding.
- Explain the kinds of pattern this course looks for.
- Set up a study so its findings can be defended.

1 The two problems

FINDING CANDIDATE STRUCTURE EFFICIENTLY, AND DECIDING WHETHER IT IS REAL, ARE DIFFERENT PROBLEMS

The algorithmic problem is that the data is too large for the obvious method — all-pairs comparison, a full sort, holding everything in memory, or an exact count — so you need an approximation with a stated bound (§2).

The statistical problem is that a large search space contains apparent patterns purely by chance — so finding one tells you about your search procedure at least as much as about the world (§3).

And they pull in opposite directions. Making the search more efficient lets you test many more hypotheses, which makes the second problem strictly worse — so an advance in the first problem is a liability in the second unless it is accounted for. Which is exactly why treating them separately fails.

So this course alternates between them deliberately — and Module 11 is its centre rather than a late caveat, because every technique before it generates candidates that need to survive it, and a course that taught the techniques without it would be teaching how to generate false findings efficiently.

Task	The question it asks	Module
Similar items	Which records are near-duplicates or near-neighbours?	02, with the approximation that makes it feasible.
Frequent patterns	Which combinations of items co-occur more than expected?	03
Clusters	Are there natural groups, and how would you know?	04
Low-dimensional structure	Does the data lie near a much simpler space?	05

Task	The question it asks	Module
Stream summaries	What can a single pass with bounded memory tell you?	06
Graph structure	Which nodes form communities, and which are central?	07
Anomalies	Which records do not belong, and compared to what?	08

Every row is unsupervised, which is what makes evaluation hard throughout the course — there is no label telling you whether a cluster was correct or an anomaly was genuinely anomalous. So each module has to construct its own answer to 'how would I know', and that recurring question is as much the course's content as the algorithms are.

2 What scale does

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n = 10^3    everything works. Use the exact method
            and do not be clever.

n = 10^6    O(n^2) is 10^12 operations -- hours to
            days. Approximation becomes attractive.

n = 10^9    O(n^2) is impossible. O(n log n) is fine.
            The data does not fit in memory, so the
            number of PASSES matters more than the
            arithmetic does.

n = 10^12   one pass, bounded memory, distributed.
            Many exact answers are unavailable at any
            price.

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AND THE BINDING CONSTRAINT IS USUALLY NOT ARITHMETIC
memory, because the working set must fit
disk and network, because a pass over the data
costs more than the computation performed
on it
and random access, which is catastrophic on data
that is not in memory

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WHICH IS WHY sequential single-pass algorithms win at
scale even when they perform more arithmetic.

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Passes over the data, not operations, is the right cost model once the data exceeds memory — and that reframes the whole course: it is why the streaming material of Module 06 is practical rather than exotic, why Module 03's algorithms are organised around pass count, and why Module 10's distributed systems care about shuffles. **CSCE 629's complexity analysis is still correct and is answering a question that is no longer the binding one.**

SO APPROXIMATION IS NOT A COMPROMISE; IT IS FREQUENTLY THE ONLY AVAILABLE ANSWER

An approximate answer with a stated error bound is a real result — and **an exact answer you cannot compute is not an answer at all**, which is the framing to adopt before the approximation starts feeling like a concession.

Which is exactly why CSCE 658 is a prerequisite rather than a recommendation: hashing, sampling, and concentration bounds are what let you say *how* approximate your answer is, and an approximation without a bound is simply a guess with no claim attached.

And the bound has to be verified empirically. Theoretical guarantees are asymptotic and assume things about your data — independence, a hash function's behaviour, a distribution's tails — **all of which are worth checking** rather than assuming.

So the discipline is: choose the approximation, state the bound, then measure whether the bound holds on your data — which is **Project 1's central requirement** and is the habit that distinguishes using these methods from invoking them.

3 Why search finds things

$$E[\text{false positives}] = m\alpha \quad P(\text{at least one}) = 1 - (1-\alpha)^m$$

- Test m independent hypotheses at level α , and the expected number of false positives is $m\alpha$ — regardless of whether any real effect exists anywhere.
- So $m = 10,000$ at $\alpha = 0.05$ gives 500 false findings in pure noise — and every single one of them has $p < 0.05$, which is the whole difficulty: they are indistinguishable from real findings by the statistic alone.
- And the probability of at least one false positive is $1 - (1-\alpha)^m$, which is 0.9999... for any large m — so finding something is not merely likely but effectively guaranteed.
- Which means the finding carries no information until the search is accounted for. This equation is the course's moral centre, and it is stated in Module 01 rather than Module 11 because every technique in between has to be read with it in mind.

AND DATA MINING SEARCHES ENORMOUS HYPOTHESIS SPACES BY CONSTRUCTION

Frequent itemset mining over a thousand items considers 2^{1000} candidate sets in principle — and reports whichever ones pass a support threshold, which is a search over an astronomically large space conducted on your behalf by the algorithm (Module 03).

Correlation mining over ten thousand variables tests approximately fifty million pairs — so at the five percent level, two and a half million pairs look significant with no structure in the data at all, and the top of that list looks extremely compelling.

And the searches are not independent, which makes the correction harder rather than unnecessary — the naive Bonferroni bound remains valid and becomes very conservative, and Module 11 §2's methods exist to do better.

So the field's central honesty requirement is counting the hypotheses — and the count is the number everybody understates, because it includes the analyses you abandoned, the parameters you tried, and the preprocessing choices you made, all of which were hypothesis tests whether or not you recorded them (Module 11 §1).

4 Setting up to be believed

- Split off a validation portion and do not touch it. A pattern that survives data you had not seen when you found it is a categorically different kind of evidence from one found in all of the data — and the split has to happen first.
- Write down what you expect to find. A predicted pattern confirmed is far stronger evidence than an unpredicted one found, because the prediction fixed the hypothesis count at one — and the record has to predate the search to mean anything.
- Keep a log of every hypothesis tested, including the abandoned ones, the parameter sweeps, and the alternative preprocessing — because Module 11 needs the count and it genuinely cannot be reconstructed afterwards from memory or from the notebook.

- **Decide in advance what effect size would actually matter** — which **prevents a statistically significant and practically useless finding from counting as a result**, a failure mode that large n makes near-certain (Module 11 §3).
- **And work out in advance what would explain the pattern more cheaply**: a confounder, a sampling artefact, or something about how the data was collected. **The hypothesis log is the cheapest of these five and the one nobody keeps** — and **without it, Module 11's correction cannot be applied honestly at all**, which makes everything downstream unverifiable.

HOW TO READ THIS COURSE

Modules 02 to 08 are the techniques, and each one comes with three things: **its approximation, the bound on that approximation, and its evaluation problem** — which is the pattern to look for in each.

Modules 09 and 10 are application and scale — recommendation as the worked end-to-end system, and what distribution actually costs once you stop assuming it is free.

Module 11 is the centre, and **every technique before it produces candidates that have to pass it** — so it is worth reading its first section early and returning to it.

And Modules 12 and 13 are about the people in the data and about honest claims — **because mining data about people has consequences for those people**, and **a found pattern is a claim about a search procedure** rather than about the world, until the work in Module 11 has been done.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Mining of Massive Datasets, chapter 1 (free)	§1 through §3 — and its section on 'bonferroni's principle' is this course's §3, stated by the authors with a memorable example.
Ioannidis — Why Most Published Research Findings Are False (free)	§3's argument with the arithmetic — short, and it changed how several fields report results.
Simmons, Nelson & Simonsohn — False-Positive Psychology (free)	§4's motivation — how ordinary analytic flexibility produces significant findings from nothing.
Stanford CS246, lecture 1 (free)	§2's scale thresholds , with the cost model argued concretely.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **State the two problems** and explain how an improvement in one worsens the other.
- 2 **Compute the operation count** for all-pairs comparison at a million and a billion records.
- 3 **Time a single sequential pass** over a large file and compare against random access over the same data.
- 4 **Generate pure noise with 10,000 variables** and count the pairs with $p < 0.05$.
- 5 **Report the most significant one** and note how compelling it looks.
- 6 **Compute the probability of at least one false positive** for $m = 100, 10,000$, and 10 million.
- 7 **Count the candidate itemsets** over 1,000 items.
- 8 **Split a dataset into exploration and validation**, and write down your predictions before looking.
- 9 **Start a hypothesis log** and commit to it for the semester.
- 10 **Write down three cheaper explanations** for a pattern you expect to find.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 State the two problems and why they pull in opposite directions.
- 2 Why is Module 11 the centre rather than a caveat?
- 3 Name seven tasks and say what they have in common.
- 4 What is the right cost model above memory size, and why?
- 5 Why is approximation not a compromise?
- 6 Why must a bound be verified empirically?
- 7 Give the multiplicity arithmetic and what it implies.
- 8 Why does data mining have this problem worst?
- 9 Give five things to do before the first query.
- 10 Which is cheapest, and why does its absence break everything downstream?